HW5 Kagan Ucak

1- $\omega(z) = \Delta\omega = \omega_0 - \omega \rightarrow \dfrac{dM_r}{dt} = i\,\omega(z) M_r + i\,\omega_1(t) M_0$

$$\hookrightarrow M_r(t,z) = i M_0 e^{-i\omega t} \int_0^t e^{i\omega s}\,\omega_1\,ds$$

nonselective

↳ gradient = 0

↳ $\omega_1(t) = \gamma B_1 \xrightarrow{\text{at } t=\tau} M_r(\tau) = i M_0 e^{-i\Delta\omega\tau} \int_0^\tau e^{i\Delta\omega s}\,\gamma B_1\,ds$

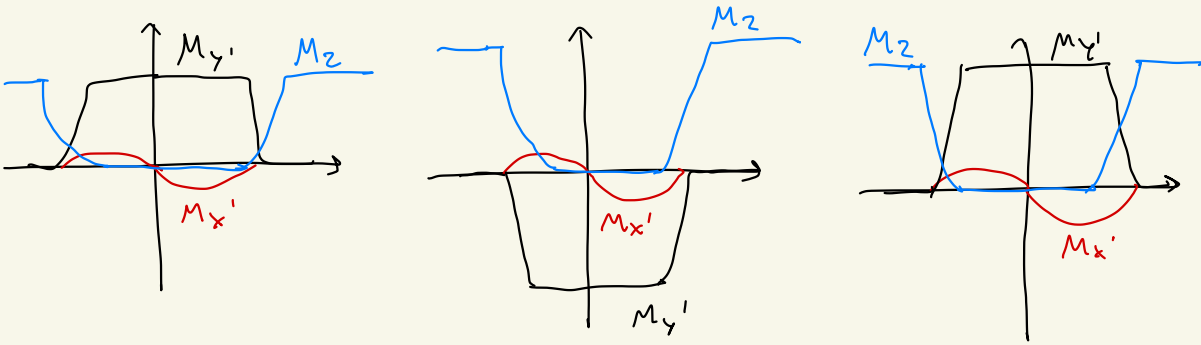
$$= i M_0 e^{-i\Delta\omega\tau}\,\gamma B_1 \int_0^\tau e^{i\Delta\omega s}\,ds$$

$$= \cancel{i}\gamma B_1 M_0 e^{-i\Delta\omega\tau}\left(\frac{e^{i\Delta\omega\tau} - 1}{\cancel{i}\Delta\omega}\right)$$

$$\boxed{M_r(\tau) = \gamma B_1 M_0 \frac{(1 - e^{-i\Delta\omega\tau})}{\Delta\omega}}$$

$$\boxed{|M_r(\tau)| = \gamma B_1 M_0 \frac{|(1 - e^{-i\Delta\omega\tau})|}{\Delta\omega}}$$

2- a) b) c)

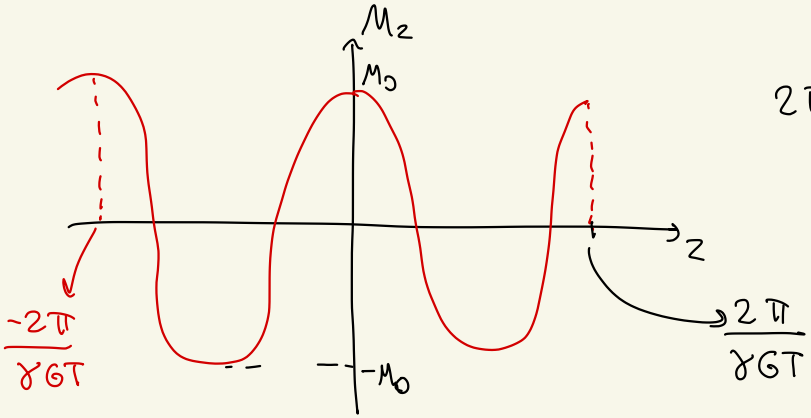


Yes, results would still hold true if the approximation was not assumed.

3-) After 90° x' pulse $\rightarrow M = (0, M_0, 0)$

After gradient $\rightarrow M = (-M_0 \sin(\phi(z)), M_0 \cos(\phi(z)), 0)$

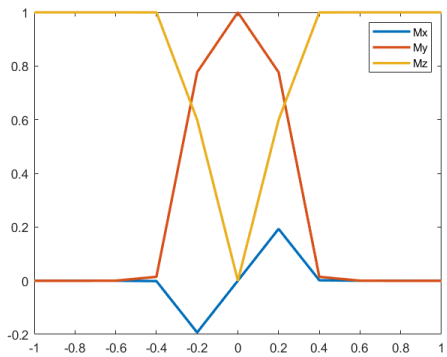
After -90° x' pulse $\rightarrow M_z(z) = M_0 \cdot \cos(\gamma G T z)$

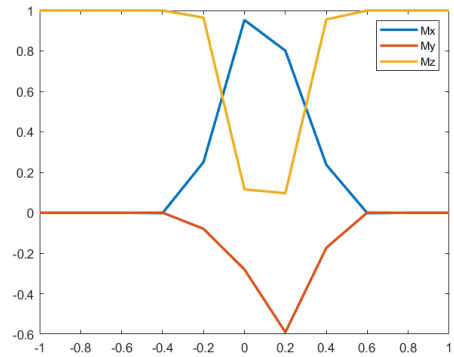


$2\pi = \gamma G T z_1$

$z_1 = \frac{2\pi}{\gamma G T}$

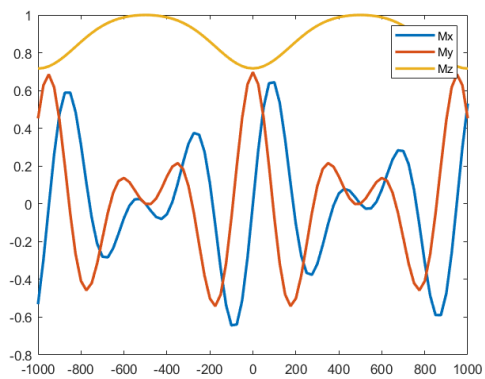
Part A

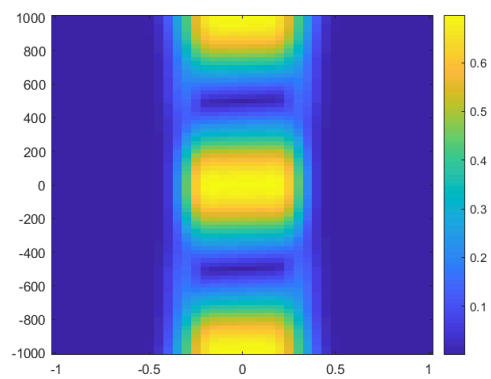
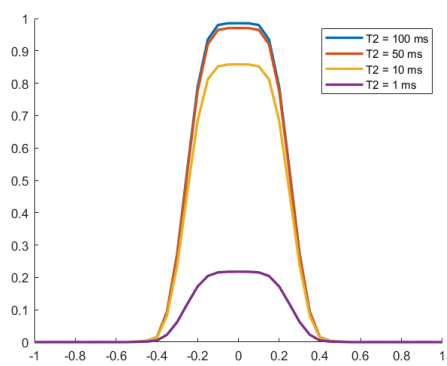




X profiles changed a lot, and it looks like the Y profile in the previous setting. Y's profile changed a lot, too. Its sign flipped, and the peak magnitude decreased to -0.6. The Z profile has slightly changed and does not touch the 0 value now.
We will only see the fat as shining that overlaps with the excitation frequency of the fat.

Part B





Part C